transactions are executed.

Now imagine that instead of storing the account balances, we store only the transactions. Whenever anyone wants to know the balance of an account, we simply add up all the transactions for that account, from the beginning of time. This scheme requires no mutable variables.

Obviously, this approach sounds absurd. Over time, the number of transactions would grow without bound, and the processing power required to compute the totals would become intolerable. To make this scheme work forever, we would need infinite storage and infinite processing power.

But perhaps we don't have to make the scheme work forever. And perhaps we have enough storage and enough processing power to make the scheme work for the reasonable lifetime of the application.

This is the idea behind *event sourcing*.[2] Event sourcing is a strategy wherein we store the transactions, but not the state. When state is required, we simply apply all the transactions from the beginning of time.

Of course, we can take shortcuts. For example, we can compute and save the state every midnight. Then, when the state information is required, we need compute only the transactions since midnight.

Now consider the data storage required for this scheme: We would need a lot of it. Realistically, offline data storage has been growing so fast that we now consider trillions of bytes to be small—so we have a lot of it.

More importantly, nothing ever gets deleted or updated from such a data store. As a consequence, our applications are not CRUD; they are just CR. Also, because neither updates nor deletions occur in the data store, there cannot be any concurrent update issues.

If we have enough storage and enough processor power, we can make our applications entirely immutable—and, therefore, *entirely functional*.

If this still sounds absurd, it might help if you remembered that this is precisely the way your source code control system works.

CONCLUSION

To summarize:

- Structured programming is discipline imposed upon direct transfer of control.
- Object-oriented programming is discipline imposed upon indirect transfer of control.
- Functional programming is discipline imposed upon variable assignment.

Each of these three paradigms has taken something away from us. Each restricts some aspect of the way we write code. None of them has added to our power or our capabilities.

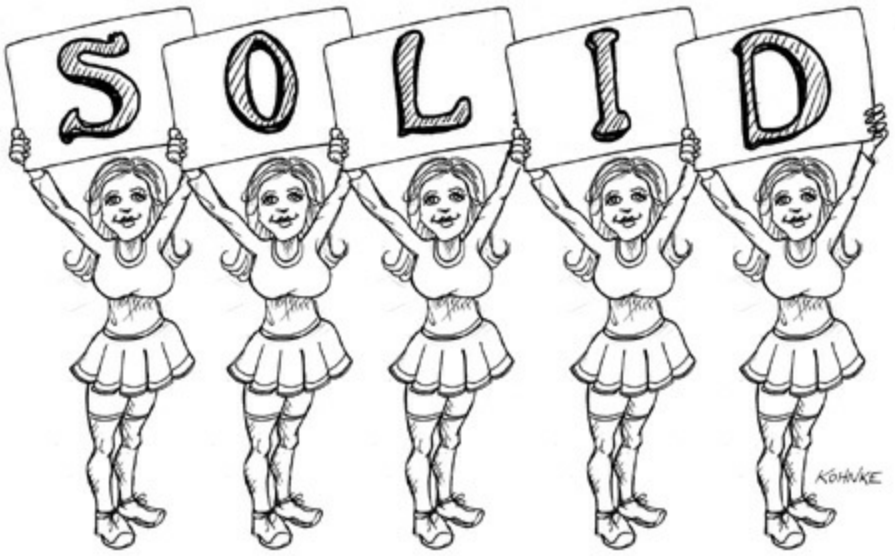
What we have learned over the last half-century is *what not to do*.

With that realization, we have to face an unwelcome fact: Software is not a rapidly advancing technology. The rules of software are the same today as they were in 1946, when Alan Turing wrote the very first code that would execute in an electronic computer. The tools have changed, and the hardware has changed, but the essence of software remains the same.

Software—the stuff of computer programs—is composed of sequence, selection, iteration, and indirection. Nothing more. Nothing less.

1. I know... What's a disk?
2. Thanks to Greg Young for teaching me about this concept.

III
DESIGN PRINCIPLES



Good software systems begin with clean code. On the one hand, if the bricks aren't well made, the architecture of the building doesn't matter much. On the other hand, you can make a substantial mess with well-made bricks. This is where the SOLID principles come in.

The SOLID principles tell us how to arrange our functions and data structures into classes, and how those classes should be interconnected. The use of the word "class" does not imply that these principles are applicable only to object-oriented software. A class is simply a coupled grouping of functions and data. Every software system has such groupings, whether they are called classes or not. The SOLID principles apply to those groupings.

The goal of the principles is the creation of mid-level software structures that:

- Tolerate change,
- Are easy to understand, and
- Are the basis of components that can be used in many software systems.

The term “mid-level” refers to the fact that these principles are applied by programmers working at the module level. They are applied just above the level of the code and help to define the kinds of software structures used within modules and components.

Just as it is possible to create a substantial mess with well-made bricks, so it is also possible to create a system-wide mess with well-designed mid-level components. For this reason, once we have covered the SOLID principles, we will move on to their counterparts in the component world, and then to the principles of high-level architecture.

The history of the SOLID principles is long. I began to assemble them in the late 1980s while debating software design principles with others on USENET (an early kind of Facebook). Over the years, the principles have shifted and changed. Some were deleted. Others were merged. Still others were added. The final grouping stabilized in the early 2000s, although I presented them in a different order.

In 2004 or thereabouts, Michael Feathers sent me an email saying that if I rearranged the principles, their first words would spell the word SOLID—and thus the SOLID principles were born.

The chapters that follow describe each principle more thoroughly. Here is the executive summary:

- **SRP:** The Single Responsibility Principle

  An active corollary to Conway’s law: The best structure for a software system is heavily influenced by the social structure of the organization that uses it so that each software module has one, and only one, reason to change.
- **OCP:** The Open-Closed Principle

  Bertrand Meyer made this principle famous in the 1980s. The gist is that for software systems to be easy to change, they must be designed to allow the behavior of those systems to be changed by adding new code, rather than changing existing code.
- **LSP:** The Liskov Substitution Principle

  Barbara Liskov’s famous definition of subtypes, from 1988. In short, this principle

says that to build software systems from interchangeable parts, those parts must adhere to a contract that allows those parts to be substituted one for another.

- **ISP:** The Interface Segregation Principle

  This principle advises software designers to avoid depending on things that they don't use.

- **DIP:** The Dependency Inversion Principle

  The code that implements high-level policy should not depend on the code that implements low-level details. Rather, details should depend on policies.

These principles have been described in detail in many different publications[1] over the years. The chapters that follow will focus on the architectural implications of these principles instead of repeating those detailed discussions. If you are not already familiar with these principles, what follows is insufficient to understand them in detail and you would be well advised to study them in the footnoted documents.

1. For example, *Agile Software Development, Principles*, Patterns, and Practices, Robert C. Martin, Prentice Hall, 2002, http://www.butunclebob.com/ArticleS.UncleBob.PrinciplesOfOod, and https://en.wikipedia.org/wiki/SOLID_(object-oriented_design) (or just google SOLID).

7

SRP: THE SINGLE RESPONSIBILITY PRINCIPLE

Of all the SOLID principles, the Single Responsibility Principle (SRP) might be the least well understood. That's likely because it has a particularly inappropriate name. It is too easy for programmers to hear the name and then assume that it means that every module should do just one thing.

Make no mistake, there *is* a principle like that. A *function* should do one, and only one, thing. We use that principle when we are refactoring large functions into smaller functions; we use it at the lowest levels. But it is not one of the SOLID principles—it is not the SRP.

Historically, the SRP has been described this way:

A module should have one, and only one, reason to change.

Software systems are changed to satisfy users and stakeholders; those users and stakeholders *are* the "reason to change" that the principle is talking about. Indeed, we can rephrase the principle to say this:

A module should be responsible to one, and only one, user or stakeholder.

Unfortunately, the words "user" and "stakeholder" aren't really the right words to use here. There will likely be more than one user or stakeholder who wants the system changed in the same way. Instead, we're really referring to a group—one or more people who require that change. We'll refer to that group as an *actor*.

Thus the final version of the SRP is:

A module should be responsible to one, and only one, actor.

Now, what do we mean by the word "module"? The simplest definition is just a source file. Most of the time that definition works fine. Some languages and development environments, though, don't use source files to contain their code. In those cases a module is just a cohesive set of functions and data structures.

That word "cohesive" implies the SRP. Cohesion is the force that binds together the code responsible to a single actor.

Perhaps the best way to understand this principle is by looking at the symptoms of violating it.

SYMPTOM 1: ACCIDENTAL DUPLICATION

My favorite example is the `Employee` class from a payroll application. It has three methods: `calculatePay()`, `reportHours()`, and `save()` (Figure 7.1).

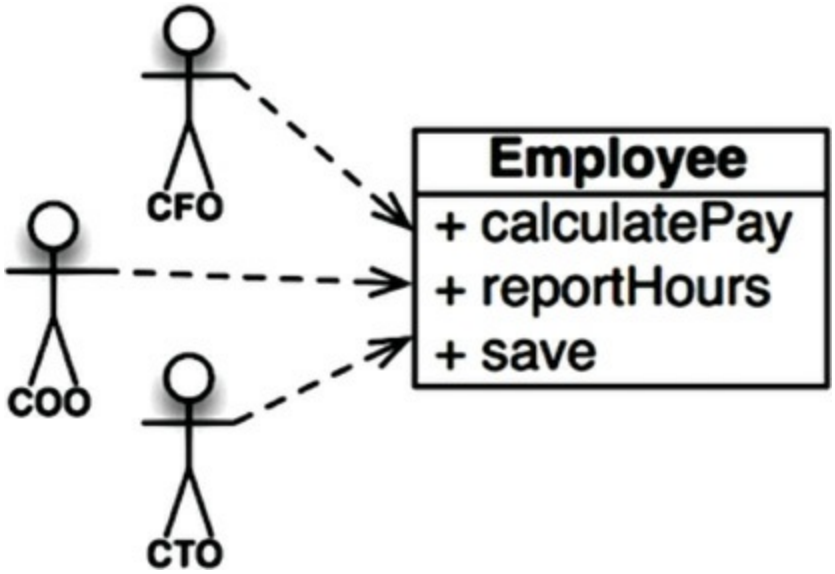


Figure 7.1 The `Employee` class

This class violates the SRP because those three methods are responsible to three very different actors.

- The `calculatePay()` method is specified by the accounting department, which reports to the CFO.
- The `reportHours()` method is specified and used by the human resources department, which reports to the COO.
- The `save()` method is specified by the database administrators (DBAs), who report to the CTO.

By putting the source code for these three methods into a single `Employee` class, the developers have coupled each of these actors to the others. This coupling can cause the actions of the CFO's team to affect something that the COO's team depends on.

For example, suppose that the `calculatePay()` function and the `reportHours()` function share a common algorithm for calculating non-overtime hours. Suppose also that the developers, who are careful not to duplicate code, put that algorithm into a function named `regularHours()` (Figure 7.2).